

Week 9 Lab

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INFT1203: Access Controls

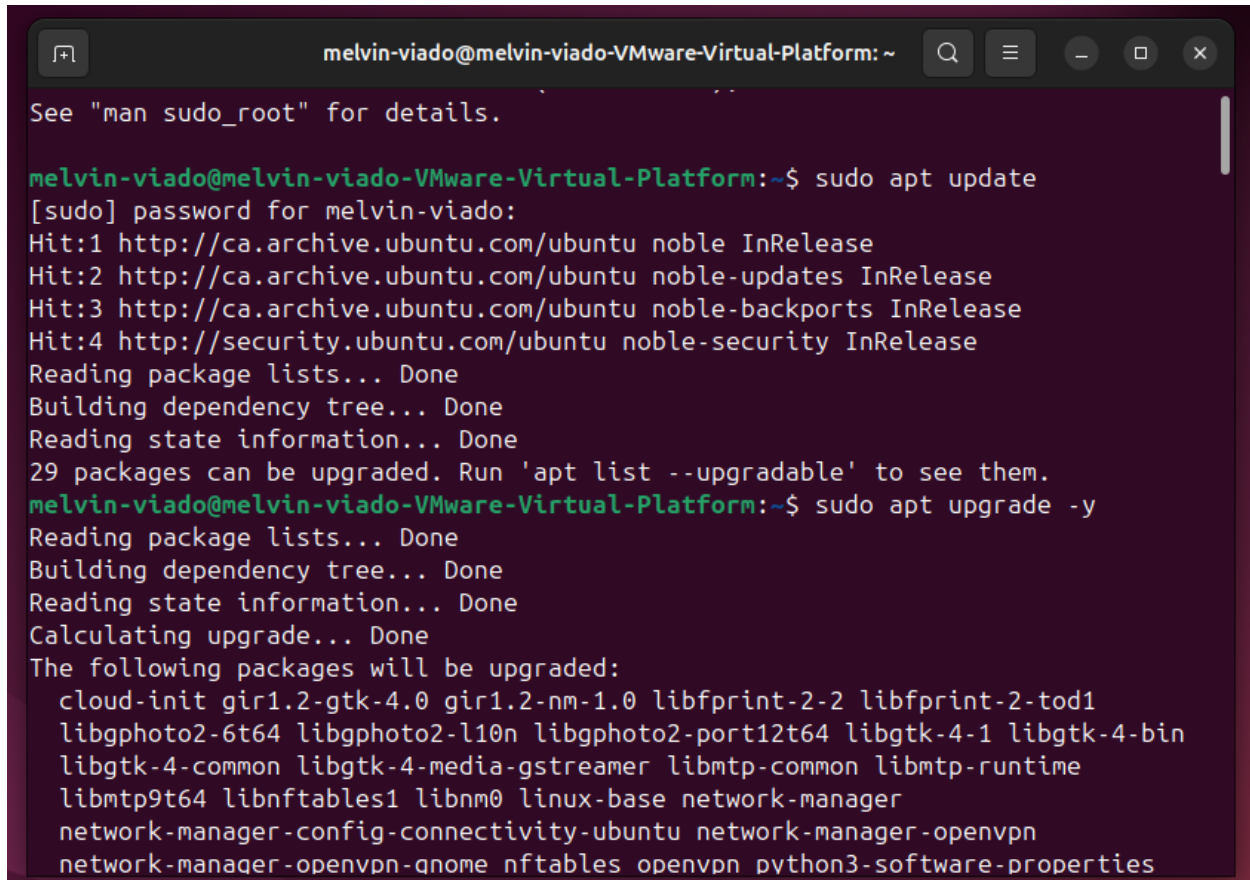
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Best Practices

Screenshots

Ensuring an up to date system

A terminal window with a dark purple background. The title bar shows the user 'melvin-viado' and the host 'melvin-viado-VMware-Virtual-Platform'. The terminal displays the output of 'sudo apt update' and 'sudo apt upgrade -y'. The first command shows four hits from Ubuntu repositories and lists 29 upgradable packages. The second command lists the specific packages to be upgraded, including cloud-init, gir1.2-gtk-4.0, gir1.2-nm-1.0, libfprint-2-2, libfprint-2-tod1, libgphoto2-6t64, libgphoto2-l10n, libgphoto2-port12t64, libgtk-4-1, libgtk-4-bin, libgtk-4-common, libgtk-4-media-gstreamer, libmtp-common, libmtp-runtime, libmtp9t64, libnftables1, libnm0, linux-base, network-manager, network-manager-config-connectivity-ubuntu, network-manager-openvpn, network-manager-openvpn-gnome, nftables, openvpn, and python3-software-properties.

```
melvin-viado@melvin-viado-VMware-Virtual-Platform: ~  
See "man sudo_root" for details.  
  
melvin-viado@melvin-viado-VMware-Virtual-Platform:~$ sudo apt update  
[sudo] password for melvin-viado:  
Hit:1 http://ca.archive.ubuntu.com/ubuntu noble InRelease  
Hit:2 http://ca.archive.ubuntu.com/ubuntu noble-updates InRelease  
Hit:3 http://ca.archive.ubuntu.com/ubuntu noble-backports InRelease  
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
29 packages can be upgraded. Run 'apt list --upgradable' to see them.  
melvin-viado@melvin-viado-VMware-Virtual-Platform:~$ sudo apt upgrade -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
Calculating upgrade... Done  
The following packages will be upgraded:  
  cloud-init gir1.2-gtk-4.0 gir1.2-nm-1.0 libfprint-2-2 libfprint-2-tod1  
  libgphoto2-6t64 libgphoto2-l10n libgphoto2-port12t64 libgtk-4-1 libgtk-4-bin  
  libgtk-4-common libgtk-4-media-gstreamer libmtp-common libmtp-runtime  
  libmtp9t64 libnftables1 libnm0 linux-base network-manager  
  network-manager-config-connectivity-ubuntu network-manager-openvpn  
  network-manager-openvpn-gnome nftables openvpn python3-software-properties
```

Install required packages

```

melvin-viado@melvin-viado-VMware-Virtual-Platform: ~
Setting up libgtk-4-media-gstreamer (4.14.5+ds-0ubuntu0.9) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
melvin-viado@melvin-viado-VMware-Virtual-Platform:~$ sudo apt install php apache2 freeradius libapache2-mod-php mariadb-
server freeradius-mysql freeradius-utils php-{gd,common,mail,mail-mime,mysql,pear,db,mbstring,xml,curl} wget unzip zip -
y

Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
wget is already the newest version (1.21.4-1ubuntu4.1).
wget set to manually installed.
unzip is already the newest version (6.0-28ubuntu4.1).
unzip set to manually installed.
zip is already the newest version (3.0-13ubuntu0.2).
zip set to manually installed.
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils freeradius-common freeradius-config
  freetds-common galera-4 gawk libapache2-mod-php8.3 libapr1t64
  libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64 libcgi-fast-perl
  libcgi-pm-perl libconfig-inifiles-perl libct4 libdbd-mysql-perl libdbi-perl
  libfcgi-bin libfcgi-perl libfcgi0t64 libfreeradius3 libhtml-template-perl
  libmariadb3 libmysqlclient21 libsigsegv2 libsnappy1v5 libsodium23
  libterm-readkey-perl liburing2 make mariadb-client mariadb-client-core
  mariadb-common mariadb-plugin-provider-bzip2 mariadb-plugin-provider-lz4
  mariadb-plugin-provider-lzma mariadb-plugin-provider-lzo
  mariadb-plugin-provider-snappy mariadb-server-core mysql-common
  php-auth-sasl php-net-smtp php-net-socket php8.3 php8.3-cli php8.3-common
  php8.3-curl php8.3-gd php8.3-mbstring php8.3-mysql php8.3-opcache
  php8.3-readline php8.3-xml pv socat
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom freeradius-krb5
  freeradius-ldap freeradius-postgresql freeradius-python3 snmp gawk-doc

```

Enable Apache2, FreeRadius

```

melvin-viado@melvin-viado-VMware-Virtual-Platform:~$ sudo systemctl enable --now apache2 && sudo systemctl enable freeradi
us

Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
Synchronizing state of freeradius.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable freeradius

```

SQL Server Setup

```
Cleaning up...
melvin-viado@melvin-viado-VMware-Virtual-Platform:~$ sudo mysql_secure_installation

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
SERVERS IN PRODUCTION USE! PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password or using the unix_socket ensures that nobody
can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n'.

Switch to unix_socket authentication [Y/n] n
... skipping.

You already have your root account protected, so you can safely answer 'n'.

Change the root password? [Y/n] n
... skipping.

By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them. This is intended only for testing, and to make the installation
go a bit smoother. You should remove them before moving into a
production environment.
```

Configure the Database

```
melvin-viado@melvin-viado-VMware-Virtual-Platform:~$ sudo mysql
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 40
Server version: 10.11.14-MariaDB-0ubuntu0.24.04.1 Ubuntu 24.04

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> █
```

Create the database

```
MariaDB [(none)]> CREATE DATABASE radius;
Query OK, 1 row affected (0.000 sec)
```

Create the user account

```
MariaDB [(none)]>
MariaDB [(none)]> CREATE USER 'radius'@'localhost' IDENTIFIED by 'PASSWORD';
Query OK, 0 rows affected (0.002 sec)
```

Grant Privileges

```
MariaDB [(none)]> GRANT ALL PRIVILEGES ON radius.* TO 'radius'@'localhost';
Query OK, 0 rows affected (0.001 sec)
```

Reload the privileges

```
MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.001 sec)
```

Quit the session

```
MariaDB [(none)]> quit;
Bye
melvin-viado@melvin-viado-VMware-Virtual-Platform:~$
```

Connect the database to FreeRadius

```
melvin-viado@melvin-viado-VMware-Virtual-Platform:~$ sudo su -

root@melvin-viado-VMware-Virtual-Platform:~#

root@melvin-viado-VMware-Virtual-Platform:~# mysql -u root -p radius < /etc/freeradius/3.0/mods-config/sql/main/mysql/schema.sql
Enter password:
root@melvin-viado-VMware-Virtual-Platform:~#

root@melvin-viado-VMware-Virtual-Platform:~# exit
logout

melvin-viado@melvin-viado-VMware-Virtual-Platform:~$ sudo ln -s /etc/freeradius/3.0/mods-available/sql /etc/freeradius/3.0/mods-enabled/
```

Provide FreeRadius with the login to the SQL server and change a few settings

```
melvin-viado@melvin-viado-VMware-Virtual-Platform:~$ sudo nano /etc/freeradius/3.0/mods-enabled/sql
```

```
dialect = "mysql"
```

```
driver = "rlm_sql_${dialect}"
```

```
mysql {
    # If any of the files below are set, TLS encryption is enabled
    #
    #     ca_file = "/etc/ssl/certs/my_ca.crt"
    #     ca_path = "/etc/ssl/certs/"
    #     certificate_file = "/etc/ssl/certs/private/client.crt"
    #     private_key_file = "/etc/ssl/certs/private/client.key"
    #     cipher = "DHE-RSA-AES256-SHA:AES128-SHA"
    #
    #     tls_required = yes
    #     tls_check_cert = no
    #     tls_check_cert_cn = no
    #
}
```

```
# Connection info:
#
server = "localhost"
port = 3306
login = "radius"
password = "PASSWORD"
```

FreeRadius config

```
melvin-viado@melvin-viado-VMware-Virtual-Platform:~$ sudo chgrp -h freerad /etc/freeradius/3.0/mods-available/sql
melvin-viado@melvin-viado-VMware-Virtual-Platform:~$ sudo chown -R freerad:freerad /etc/freeradius/3.0/mods-enabled/sql
melvin-viado@melvin-viado-VMware-Virtual-Platform:~$ sudo systemctl restart freeradius
```

Dynamic VLAN Assignment

```
GNU nano 7.2 /etc/freeradius/3.0/mods-enabled/eap *
# -*- text -*-
##
## eap.conf -- Configuration for EAP types (PEAP, TTLS, etc.)
##
##     $Id: d149707f02427234b09974effc33cb0db0dc48fb $
#####
"
```

```
use_tunneled_reply = yes
```

Install DaloRadius

```
melvin-viado@melvin-viado-VMware-Virtual-Platform:~$ wget https://github.com/lirantal/daloradius/archive/1.3.zip
--2026-03-17 19:08:40-- https://github.com/lirantal/daloradius/archive/1.3.zip
Resolving github.com (github.com)... 140.82.112.3
Connecting to github.com (github.com)|140.82.112.3|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://codeload.github.com/lirantal/daloradius/zip/refs/tags/1.3 [following]
--2026-03-17 19:08:40-- https://codeload.github.com/lirantal/daloradius/zip/refs/tags/1.3
Resolving codeload.github.com (codeload.github.com)... 140.82.113.9
Connecting to codeload.github.com (codeload.github.com)|140.82.113.9|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [application/zip]
Saving to: '1.3.zip'

1.3.zip           [          ] 18.11M  2.60MB/s   in 4.6s

2026-03-17 19:08:45 (3.96 MB/s) - '1.3.zip' saved [18986248]
```

```
melvin-viado@melvin-viado-VMware-Virtual-Platform:~$ unzip 1.3.zip
Archive: 1.3.zip
9e573b8e6ee3d9483dc348a9b53cdde546391a8b
creating: daloradius-1.3/
```

```
melvin-viado@melvin-viado-VMware-Virtual-Platform:~$ cd daloradius-1.3
melvin-viado@melvin-viado-VMware-Virtual-Platform:~/daloradius-1.3$
```

```
GNU nano 7.2 /var/www/html/daloradius/library/daloradius.conf.php *
c?php /*
*****
dalorADIUS - RADIUS Web Platform * Copyright (C) 2007 - Liran Tal <liran@enginx.com> All Rights Reserved. * *
This program is free software; you can redistribute it and/or * modify it under the terms of the GNU General
Public License * as published by the Free Software Foundation; either version 2 * of the License, or (at your
option) any later version. * * You should have received a copy of the GNU General Public License * along with
this program; if not, write to the Free Software * Foundation, Inc., 59 Temple Place - Suite 330, Boston, MA
02111-1307, USA. *
*****
Description: * dalorADIUS Configuration File * * Modification Date: * Mon Mar 28 22:55:00 EDT 2011
***** */

include (dirname(__FILE__).'/version.php');

$configValues['FREERADIUS_VERSION'] = '2';
$configValues['CONFIG_DB_ENGINE'] = 'mysqli';
$configValues['CONFIG_DB_HOST'] = 'localhost';
$configValues['CONFIG_DB_PORT'] = '3306';
$configValues['CONFIG_DB_USER'] = 'radius';
$configValues['CONFIG_DB_PASS'] = 'PASSWORD';
$configValues['CONFIG_DB_NAME'] = 'radius';
$configValues['CONFIG_DB_TBL_RADCHECK'] = 'radcheck';
$configValues['CONFIG_DB_TBL_RADREPLY'] = 'radreply';
$configValues['CONFIG_DB_TBL_RADGROUPREPLY'] = 'radgroupreply';
$configValues['CONFIG_DB_TBL_RADGROUPCHECK'] = 'radgroupcheck';
$configValues['CONFIG_DB_TBL_RADUSERGROUP'] = 'radusergroup';
$configValues['CONFIG_DB_TBL_RADNAS'] = 'nas';
$configValues['CONFIG_DB_TBL_RADHG'] = 'radhuntgroup';
File Name to Write: /var/www/html/daloradius/library/daloradius.conf.php
```

```
melvin-viado@melvin-viado-VMware-Virtual-Platform:~$ sudo systemctl restart freeradius.service apache2
```

Access Page

← → ↻ Not Secure http://172.10.0.132/daloradius/index.php ☆ ⓘ ≡



RADIUS Management, Reporting, Accounting and Billing by Liran Tal

Welcome, **administrator**.
Location: **default**.

[Home](#) [Management](#) [Reports](#) [Accounting](#) [Billing](#) [GIS](#) [Graphs](#) [Config](#) [Help](#)

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